

# Differentiate your Objective

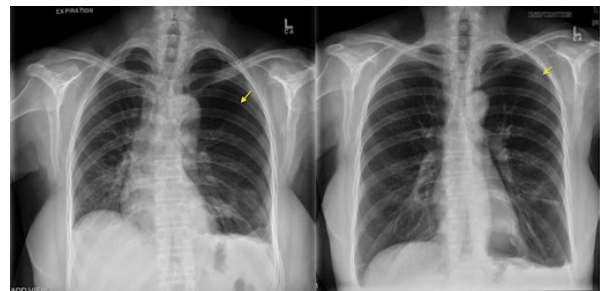
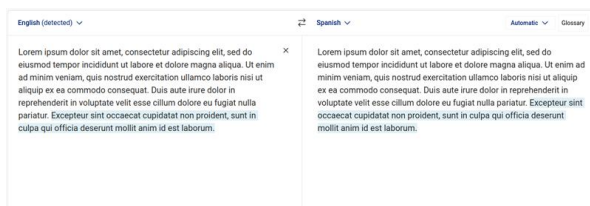
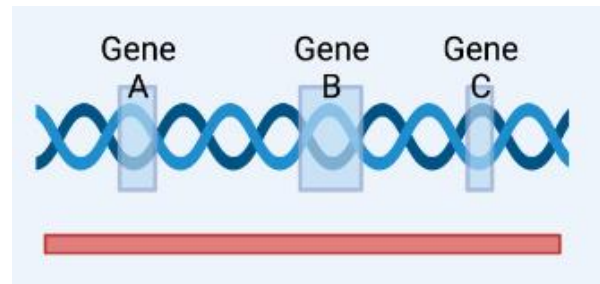
VLC =  $\int \int \int$  Vision,  
Learning & Control

AICE3002 Introduction to Deep Learning  
AICE6001 Introduction to Deep Learning (MSc)

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# What does Deep Learning (and Differentiable Programming) give us?



## Machine Learning - A Recap

All credit for this slide goes to Niranjan

Data  $\{\mathbf{x}_n, \mathbf{y}_n\}_{n=1}^N \quad \{\mathbf{x}_n\}_{n=1}^N$

Function Approximator  $\mathbf{y} = f(\mathbf{x}; \boldsymbol{\theta})$

Parameter Estimation  $E_0 = \sum_{n=1}^N \{\|\mathbf{y}_n - f(\mathbf{x}_n; \boldsymbol{\theta})\|\}^2$

Prediction  $\hat{\mathbf{y}}_{N+1} = f(\mathbf{x}_{N+1}, \hat{\boldsymbol{\theta}})$

# What is Deep Learning?

Deep learning is primarily characterised by function compositions:

- Feedforward networks:  $\mathbf{y} = f(g(\mathbf{x}; \theta_g); \theta_f)$ 
  - Often with relatively simple functions (e.g.  $f(\mathbf{x}; \theta_f) = \sigma(\mathbf{x}^\top \theta_f)$ )
- Recurrent networks:  
 $\mathbf{y}_t = f(\mathbf{y}_{t-1}, \mathbf{x}_t; \theta) = f(f(\mathbf{y}_{t-2}, \mathbf{x}_{t-1}; \theta), \mathbf{x}_t; \theta) = \dots$

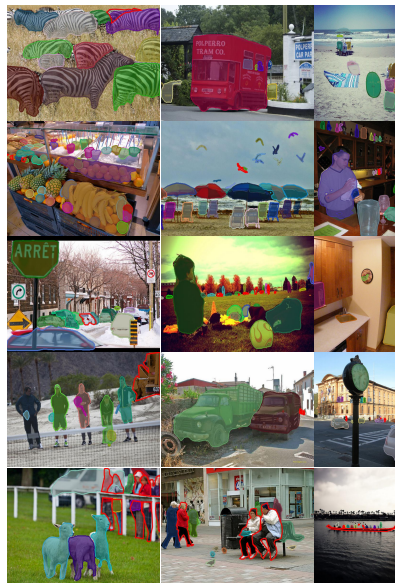
# What is Differentiable Programming?

- Captures the idea that computer programs can be partially learned rather than fully programmed by humans.
- The program is made of functions with optimisable parameters.
- So we need to be able to compute gradients with respect to the parameters of these functional blocks.
- The functional blocks don't need to be direct functions in a mathematical sense; more generally they can be *algorithms*.
- What if the functional block we're learning parameters for is itself an algorithm that optimises the parameters of an internal algorithm using a gradient based optimiser?!<sup>1</sup>

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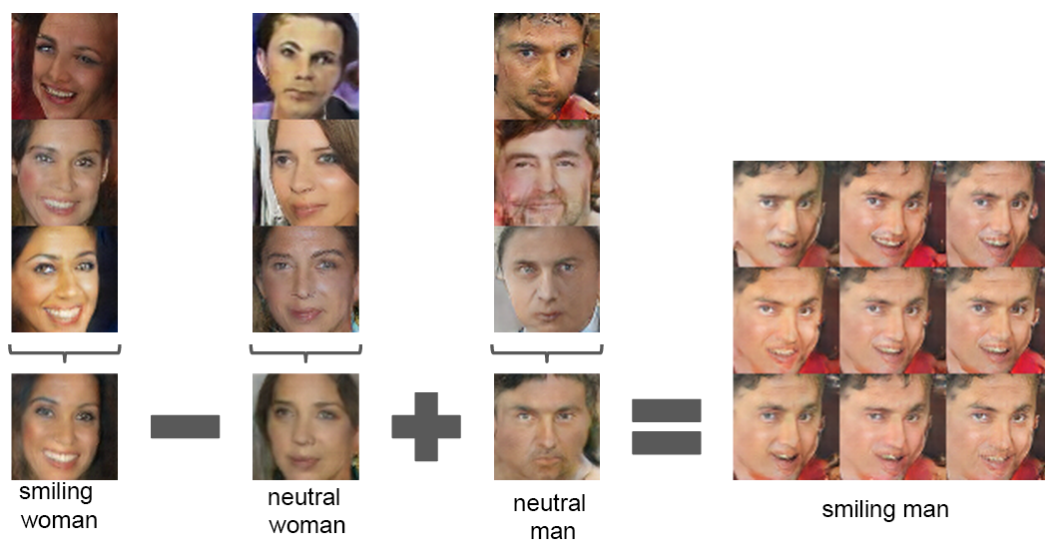
<sup>1</sup>See our ICLR 2019 paper: <https://arxiv.org/abs/1812.03928> and NeurIPS 2019 paper: <https://arxiv.org/abs/1906.06565>

## Success stories - Object detection and segmentation



Pinheiro, Pedro O., et al. "Learning to refine object segments." European Conference on Computer Vision. Springer, 2016.

## Success stories - Image generation



Radford, Alec, Luke Metz, and Soumith Chintala. "Unsupervised representation learning with deep convolutional generative adversarial networks." arXiv preprint arXiv:1511.06434 (2015).

## GPT-4

### Prompt:

Translate the following Irish text to English:

*"Bhí mé ag siúl sa choill nuair a chuala mé ceol álainn ón adharc."*

### Output (With LLM Integration):

*I was walking in the forest when I heard beautiful music from the horn.*

### Output (Traditional MT):

*I was in the wood when I heard nice sound from the horn.*

Lyu, C., Du, Z., Xu, J., Duan, Y., Wu, M., Lynn, T., Aji, A.F., Wong, D.F., Liu, S. and Wang, L., 2023. A Paradigm Shift: The Future of Machine Translation Lies with Large Language Models. LREC-COLING 2024

## Why should we care about this?

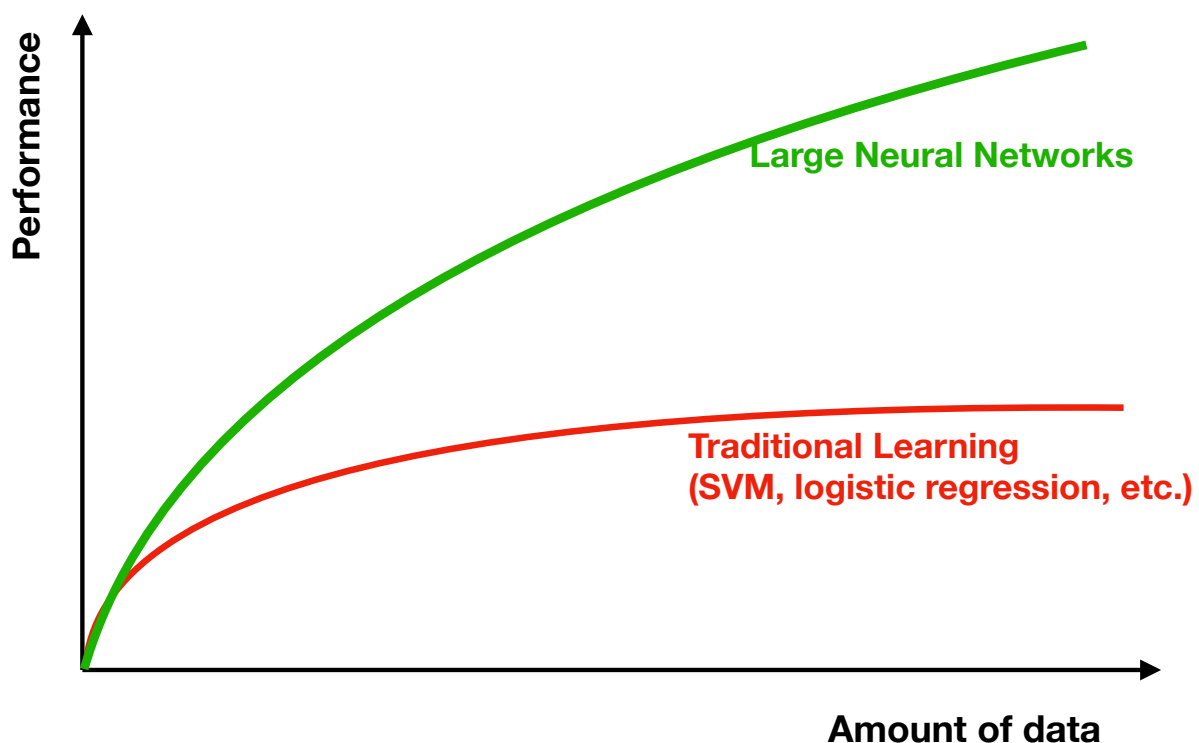


Image courtesy of Andrew Ng

# Key takeaways

- **More data availability** in all domains...
- ...allows for **bigger/deeper/more complex models**
- ...models that can **learn the features**
- and these can have far **greater performance**

## What is the objective of this module?

**A word of warning: This is not a module about how to apply someone else's deep network architecture to a task, or how to train existing models!**

You will learn some of that along the way of course, but the real objective is for you to graduate understanding how deep learning works beneath the hood.

# What is the objective of this module?

- Understand the underlying mathematical and algorithmic principles of deep learning.
- Apply deep learning methods to real datasets.
- Learn how to train models using deep learning libraries.
- Think critically about why certain approaches were designed in the way they were.

# How is this module going to be delivered?

- Lectures and seminars (3 most weeks)
  - Note: We are refreshing some material from last year's module, but the website may have old links.
  - You need to read the suggested papers/links before the lectures!
  - There is maybe a little room for some flexibility later in the course on topics - tell us what you're interested in!
  - Lectures will be face to face, but also recorded for the website when possible (do not rely on this!).

## How is this module going to be delivered?

- Seminars

- We'll look at a particular paper, papers or ideas in some detail and have an open discussion
- You'll need to prepare in advance & be ready to ask questions and share your thoughts
- Not recorded, but contents examinable

## How is this module going to be delivered?

- Labs (1x 2 hour session per week for 8 weeks + one additional optional help / catchup session)
  - Labs consist of a number of Jupyter notebooks you will work through.
    - You'll be using PyTorch as the primary framework, with Torchbearer to help out.
    - You will need to utilise GPU-compute for the later labs (we provide Google Colab links so you can use NVidia GPUs in the cloud).
  - Labs are in-person (AICE lab in B32) with a team of PhD student demonstrators & both of us.
  - Please ask lots of questions and use this time to get help on the labs.
  - After each block of labs you will have to do a follow-up problem-sheet exercise that will be marked.

# What will we cover in the module?

<https://moodle.ecs.soton.ac.uk/course/view.php?id=469>

## Lab session plan

| Lab   | Date   | Topic                                                  |
|-------|--------|--------------------------------------------------------|
| Lab 1 | 30-Sep | Tensor programming with PyTorch                        |
| Lab 2 | 07-Oct | Automatic Differentiation                              |
| Lab 3 | 14-Oct | Optimisation                                           |
| Lab 4 | 21-Oct | Linear Models                                          |
| Lab 5 | 28-Oct | NNs with PyTorch and Torchbearer                       |
|       | 04-Nov | (Catch up / help)                                      |
| Lab 6 | 11-Nov | CNNs with PyTorch and Torchbearer                      |
| Lab 7 | 18-Nov | Using pretrained models and transfer learning          |
| Lab 8 | 25-Nov | Recurrent Networks, Sequence Prediction and Embeddings |

## What do we expect you already know?

- You have have taken Foundations of Machine Learning, or are currently taking it.
- Fundamentals of:
  - Matrix Algebra (matrix-matrix, matrix-vector and matrix-scalar operations, inverse, determinant, rank, Eigendecomposition, SVD);
  - Probability & Statistics (1st-order summary statistics, simple continuous and discrete probability distributions, expected values, etc); and,
  - Multivariable Calculus (partial differentiation, chain-rule).
- Programming in Python

## What might you already know?

- How to use a deep learning framework (Keras, Tensorflow, PyTorch)?
- How to train an existing model architecture using a GPU?
- How to perform transfer learning?
- How to perform differentiable sampling of a Multivariate Normal Distribution?

- Lab work 40% - Handins in weeks 5 (23rd Oct), 9 (20th Nov), 10 (27th Nov), 12 (11th Dec)
- Final exam 60%

## What comes after this module?

- AICE6002: Deep Learning Research
  - taught by both of us
  - focus on translating the fundamentals you learn in this module to cutting edge research
  - plus lectures on advanced architectures, generative models, etc